Final Reflection Note

What I built

For my final project I built an AI assistant for public-sector documents. A user uploads a PDF, including a scanned one, and then asks questions or creates a summary, key points and action items. Every answer shows its source file and page, and the app says clearly when the answer is not in the document.

My learning journey

This final project brings together what I learned in every assignment of the internship. Each assignment added one skill, and the final project uses all of them.

Assignment	What I built	What I learned
Assignment 2	A multi-file Python package with lists, dictionaries, functions, CSV handling and exception handling	Python basics, clean code structure and safe error handling
Assignments 3 and 4	Data analysis and charts on the Netflix Titles dataset with pandas, Seaborn and Streamlit	How to explore data, find insights and show them in a simple app
Assignment 5	A Prompt Engineering Practice Book with 24 prompts (8 PITB use cases, 3 versions each) and a comparison table	How a clear prompt changes the answer, and how to compare prompts
Assignment 6	Five public-sector AI use cases: Email Assistant, Report Summarizer, Meeting Notes Generator, Document Q&A Assistant and Idea Generator	How to test prompts on real tasks and write evaluation notes

Assignment 7	A semantic search system with all-MiniLM-L6-v2 and ChromaDB, compared with keyword search	How embeddings and cosine similarity find meaning, not only words
Assignment 8	A Professional RAG PDF Assistant with Gemini and a Gradio chat interface	How to build a full RAG pipeline with sources, a fallback reply and retry logic for Gemini errors
Assignment 9	An AI Assistant Backend API with FastAPI for health check, summarization, question answering and document upload, tested in Postman	How to turn AI features into a clean, documented backend API
Final project	The Document Intelligence Assistant for public-sector documents	How to combine all of the above into one complete, tested application

How feedback helped me

My mentor, Suleman Sir, often reminded me to write clear explanations under each heading, add a README, and show real tested examples. After Assignment 8 he suggested a configurable model name and better support for scanned PDFs. I used both ideas in the final project: the model name is now an environment variable, and scanned pages are read with OCR.

What I learned

- **How RAG really works.** I learned to split text into chunks, create embeddings, store them in ChromaDB and search them. I also learned that settings like chunk size and the similarity limit change the answer quality.
- **How to make AI answers safer.** I used a strict prompt, a fallback reply and a citation check, so the app does not invent facts or page numbers.
- **How to build a clean backend.** I used FastAPI with small services, validation, logging and clear error messages for problems like a missing OCR tool or a busy AI service.
- **How to work with scanned documents.** I used PyMuPDF for normal text and Tesseract OCR only for weak pages.

- **How to test with real data.** Small examples looked fine, but real reports showed retrieval problems that I had to debug.

What was difficult

The hardest part was retrieving the correct chunks for number-based questions and handling scanned PDF pages. I improved retrieval by refining the chunk selection process and testing it with numerical questions. For scanned pages, I used OCR with Tesseract to extract text when normal PDF extraction failed, making those pages searchable and usable for answering questions.

What I would do differently

I would write the README and automated tests earlier, and I would keep a cleaner project folder from the start. I would also plan for many documents and Urdu support from the beginning.

Thank you

I want to thank my mentor, Suleman Sir, and the PITB team for their continuous guidance, support, and valuable feedback throughout this internship.

This internship has been a great learning experience for me, helping me improve not only my technical skills but also my confidence in solving real-world problems. It gave me the opportunity to apply what I had learned and understand how AI solutions are developed in practice.

This project showed me that I can take an AI idea from a plan to a working application. I also learned the importance of testing, handling challenges, and continuously improving my work.

I'm grateful for this opportunity, and I look forward to learning more, building better AI applications, and growing as an AI developer.